

2025 Shoe Sales Performance Analysis

A data-driven look at how shoe sales performed, which markets generated the highest revenue, and where future growth opportunities may exist.

Project

2025 Shoe Sales
performance Anlysis

Audience

Business Stakeholder

Prepared by

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Tools

Microsoft Excel

What we'll discuss.

01

Project Overview.

Project Overview presents the project's background, objectives, dataset, and key questions to provide context for the analysis

02

Data Cleaning.

Handling missing values, handling outlier data, removing duplicates, correcting data types, and ensuring data quality before analysis

03

Sales Performance Analysis.

Analyzing revenue trends, transaction patterns, and key business drivers

04

Dashboard Development.

Creating interactive visualizations to analyze performance, identify trends, and support business decision-making

05

Key Insight & Recommendations

Presents key findings and provides actionable recommendations based on data analysis to support business decision-making and business growth

Project Overview.

Project Background

In 2025, Company X conducted over 1,000 shoe sales transactions. As competition in the footwear industry continues to grow, **understanding the factors that drive sales performance and revenue growth has become increasingly important.** This project analyzes sales data to evaluate sales trends, identify key revenue drivers, determine top-performing segments, and generate actionable insights to support data-driven business decision-making

Data Source

<https://www.kaggle.com/datasets/syedaeman2212/shoes-sales-dataset>

Key Questions

- 01 How is the monthly sales performance trend?
- 02 Which factors have the greatest impact on revenue performance?
- 03 Which country, channel, brands and products generate the highest revenue?
- 04 What business strategies can improve overall sales performance?

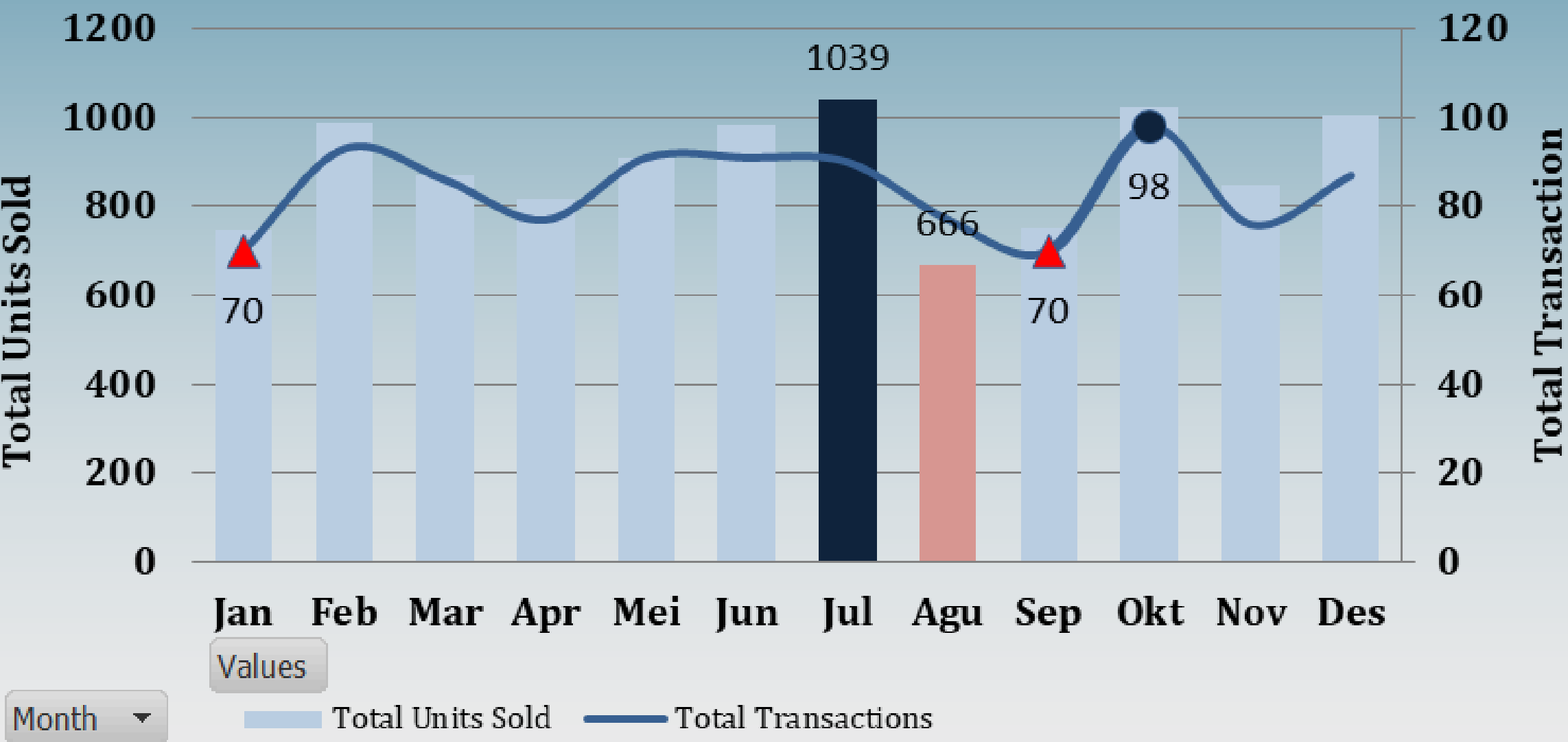
Dataset Overview

Sale_ID	: Unique transaction ID
Date	: Transaction date
Brand	: Shoe brand
Shoe_Type	: Product category
Color	: Product color
Country	: Customer location
Sales_Channel	: Sales channel
Price_USD	: Product price
Units_Sold	: Quantity sold
Revenue_USD	: Total revenue

Data Cleaning.

	<u>Data Issue</u>	<u>Action</u>	<u>Impact on Analysis</u>
01	Identify Missing Value for each variable	Removed duplicate transactions based on unique Transaction IDs	Prevents revenue overstatement and skewed counts
02	Format Inconsistencies	Standardized date formats, naming conventions (Brand/Channel), and data types.	Enables seamless aggregation and dashboard integration
03	Missing Values	Handled missing fields via category-based imputation / record removal.	Ensures 100% data completeness without introducing bias.
04	Outliers & Anomalies	Filtered extreme transaction values using standard IQR / Z-Score thresholds	Normalizes data distribution for accurate statistical testing

How do transactions trend over time?



10.651

Shoe Sold

1006

Transactions

- Similar trends pattern across months, with the exception of **June, July, and Sept**, where a divergence between the two metrics is observed
- In **June, July, and September**, total transactions declined while total units sold increased. **This pattern indicating that customers purchased more items per transaction despite lower transaction volumes**, potentially reflecting changes in purchasing behavior or basket size

 **October** recorded the highest total transaction volume and a high number of units sold, **indicating stronger customer purchasing activity during the observed period.**

 **August** recorded the lowest total units sold, accompanied by a decline in total transactions compared to the previous month. **This suggests weaker sales activity during the period.**

How do the monthly AVG revenue per transaction and product trend over time?

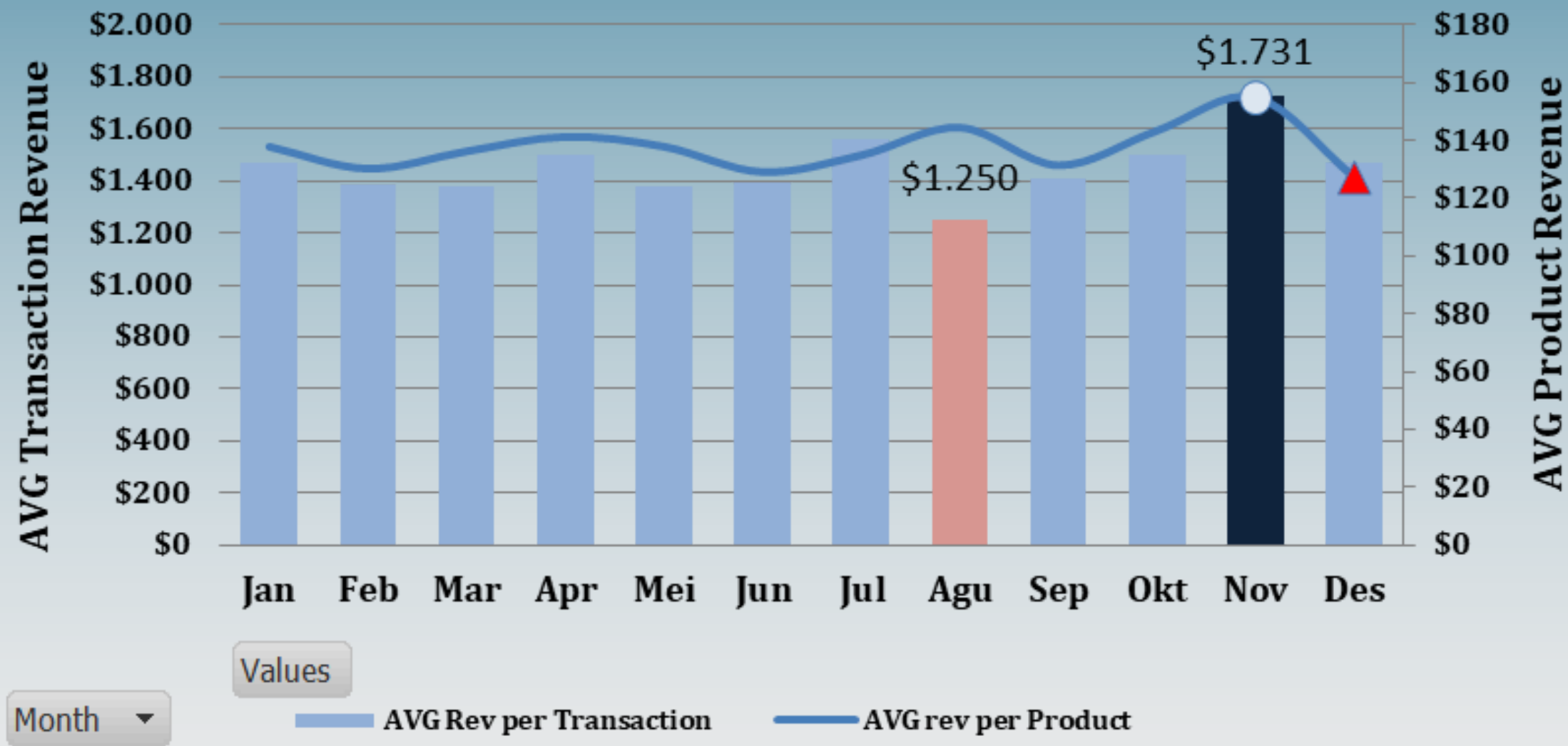
\$137

AVG Rev per Product

\$1.451

AVG Rev per Transaction

- Similar trends pattern across months, with the exception of **August and June**, where a divergence between the two metrics is observed
- In **June**, average transaction value remained stable, but average revenue per product declined, **suggesting a shift toward lower-value products despite consistent purchasing behavior**



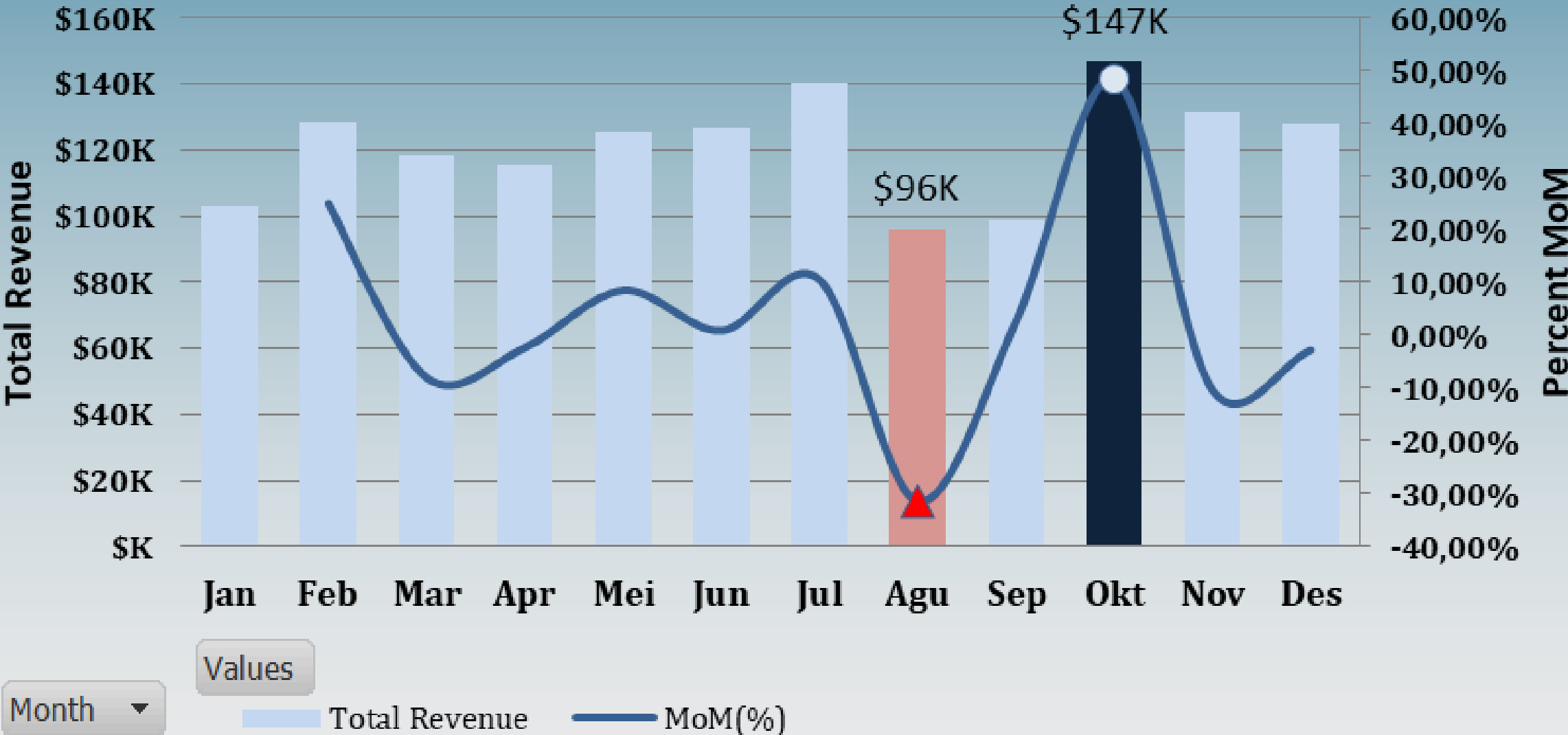
November recorded the highest across all months. However, both total transactions and units sold decreased compared to the previous month. This suggests that customers in November tended to purchase higher-value products while buying fewer items, resulting in a higher average transaction value despite lower sales volume

In August, average revenue per product was high while average transaction value was the lowest. Combined with the sharp decline in both transaction volume and units sold, this suggests that customers made fewer purchases overall but tended to purchase higher-value products

Total Revenue

\$1.459.848,63

How does revenue perform each month?



- The revenue trend and month-over-month (MoM) growth show a volatile but overall upward pattern, indicating growth with inconsistent performance. Although successive peaks increased over time, fluctuating troughs suggest that baseline performance remained unstabl
- The data also shows recovery patterns after periods of decline, with stronger rebounds but inconsistent low points.

Peak Month



October achieved the highest revenue performance This performance appears to have been driven by a combination of high transaction activity, increased sales volume, and a relatively high average revenue per transaction.

Lowest Month



August recorded the lowest total revenue and MoM growth, driven by declines across most key performance metrics, except for average revenue per product. This suggests that lower revenue performance was mainly associated with reduced sales volume and transaction activity

How do monthly sales performance trends affect business performance?

Business Concern: “Revenue Performance Variability

Although monthly revenue shows an overall upward trend, revenue and month-over-month (MoM) growth fluctuate considerably throughout the year. These **fluctuations reduce revenue predictability and make it more challenging to plan inventory, marketing activities, and business resources effectively**. Therefore, understanding the key drivers of monthly revenue performance is essential to support more stable and sustainable revenue growth

Business Question

“What are the key drivers of monthly revenue performance?”

What are the key drivers of monthly revenue performance?

Identifying the key factors that drive monthly revenue performance based on the available sales variables

1) Person Correlation Heatmap

Revenue Driver	AVG Rev per Transaction	Avg Rev per Product	Total Transactions	Total Units Sold	Total Revenue
AVG Rev per Transaction	1				
Avg Rev per Product	0,461877838	1			
Total Transactions	-0,099630766	-0,27296926	1		
Total Units Sold	0,277680416	-0,37507135	0,861187822	1	
Total Revenue	0,508216783	0,03729679	0,805175548	0,91140571	1

- **Monthly revenue performance was primarily driven by sales volume**, as total units sold showed the strongest correlation with revenue ($r = 0.91$). Additionally, total transactions also had a strong relationship with revenue ($r = 0.81$).

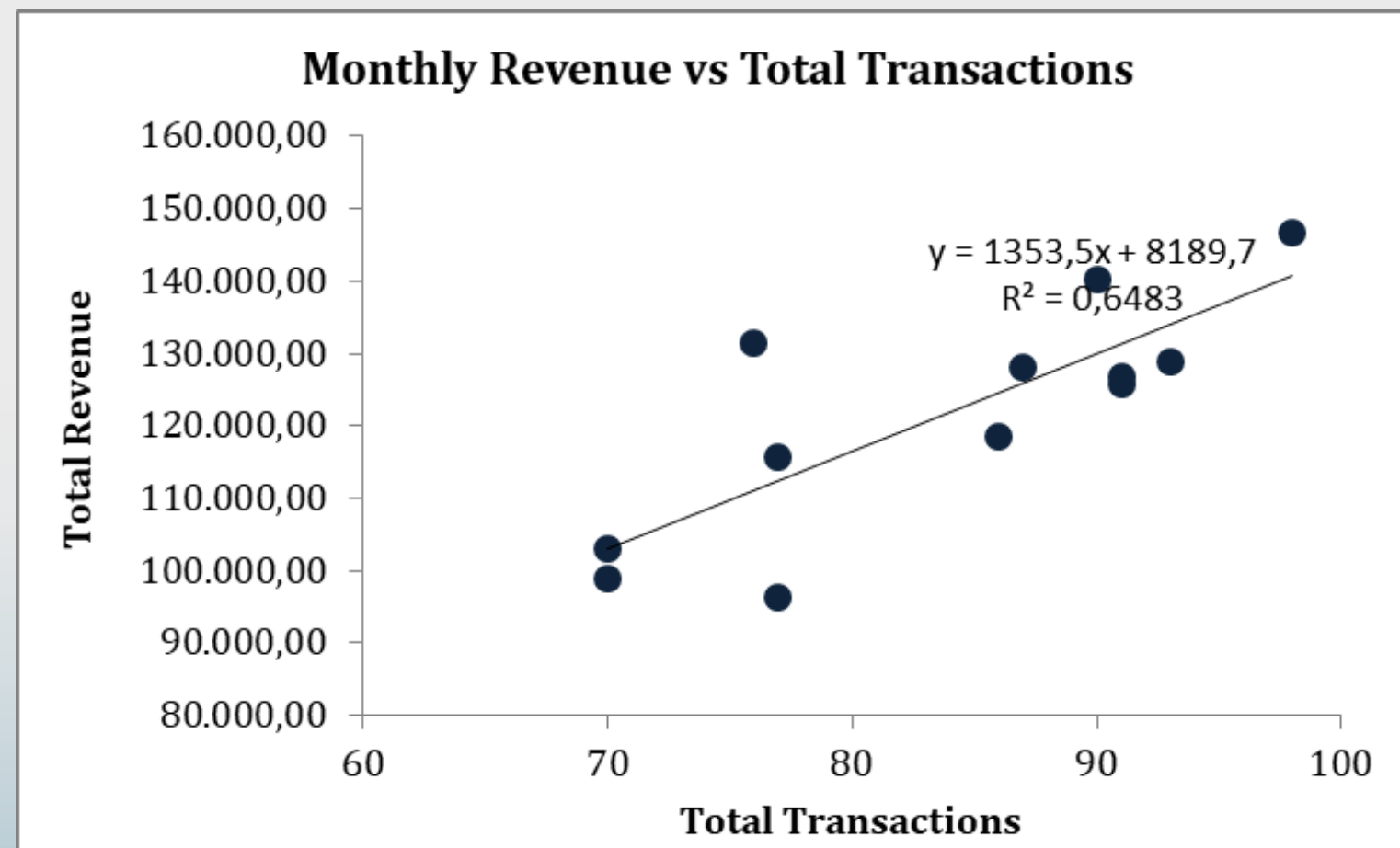
2) Driver Revenue Ranking

Rank	Revenue Driver	Person Correlation	Interpretation
1	Total Units Sold	0,911405715	Very Strong Positive Correlation
2	Total Transaction	0,805175548	Very Strong Positive Correlation
3	AVG per Transaction	0,508216783	Moderate Positive Correlation
4	AVG per Product	0,037296789	Very Weak/Negligible Correlation

- Average product value showed minimal correlation with revenue, suggesting that **revenue fluctuations were less influenced by product pricing**

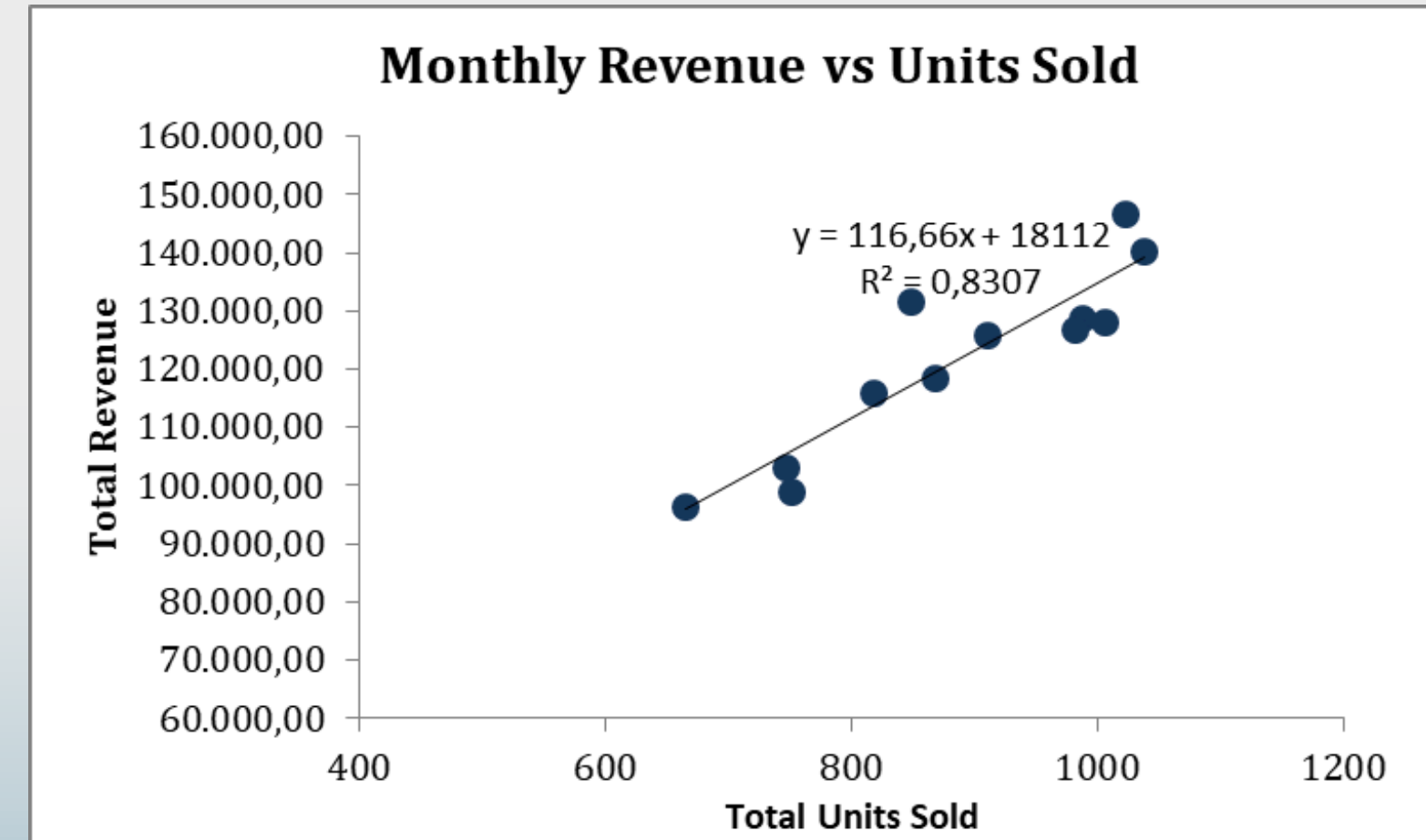
“Indicating that increasing sales volume and transaction frequency were the main contributors to revenue growth”

3) Contribution Analysis



Metric	Result
R Square	0,648307664
F-statistic	18,43394346
Coefficients	1353,451386
P-value	0,00157755

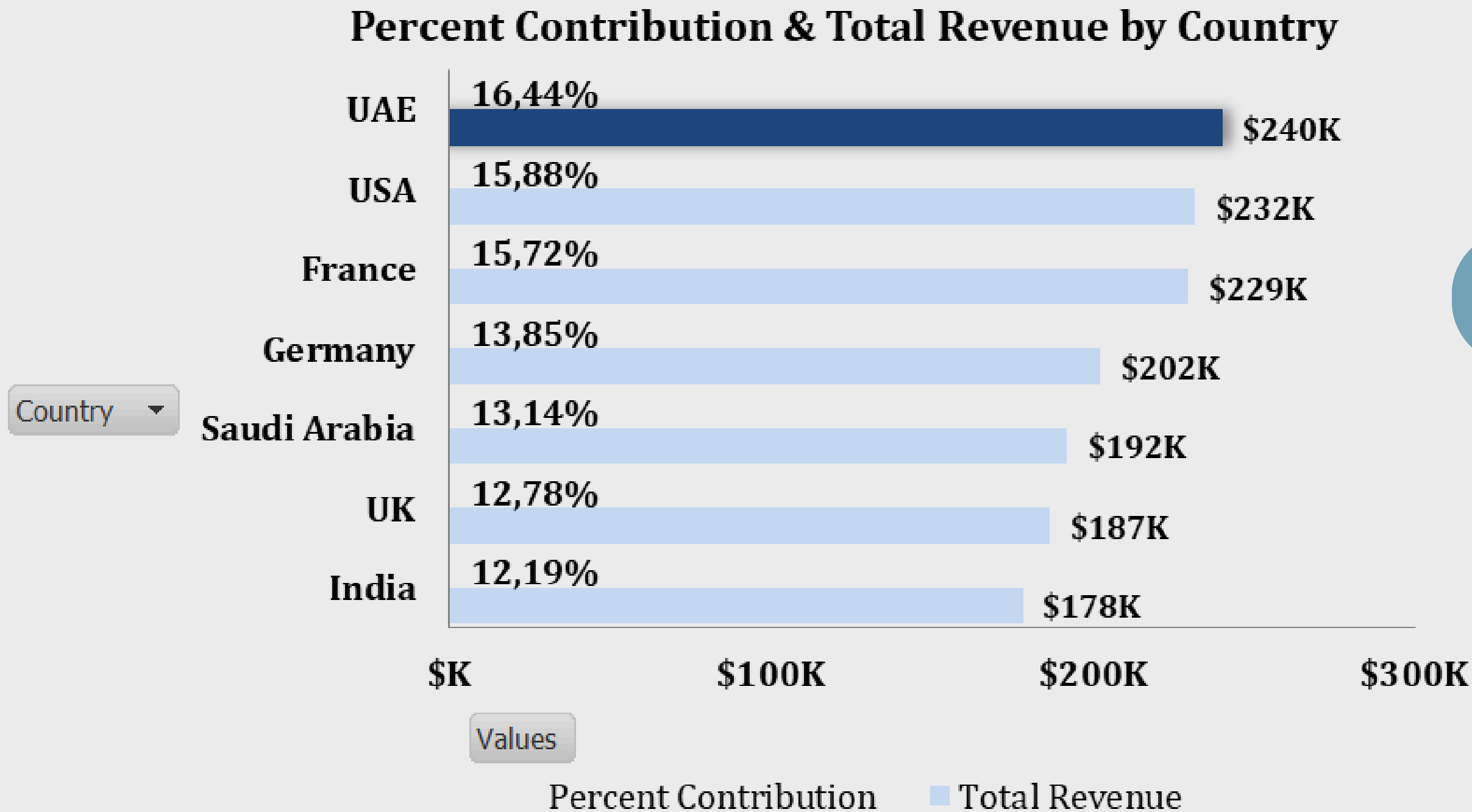
- Total Transactions is a significant driver of monthly revenue performance, explains 64,8% of the variation in monthly revenue ($F = 18,43$, $p < 0.01$).
- Each additional transaction is associated with an estimated increase of 1,353.45 in revenue.



Metric	Result
R Square	0,830660377
F-statistic	49,0529246
Coefficients	116,656425
P-value	0,00003699

- Units Sold is the strongest driver of monthly revenue performance.** The regression model explains 83.1% of the variation in monthly revenue ($F = 49.05$, $p < 0.001$).
- Each additional unit sold is associated with an estimated 116.66 increase in revenue

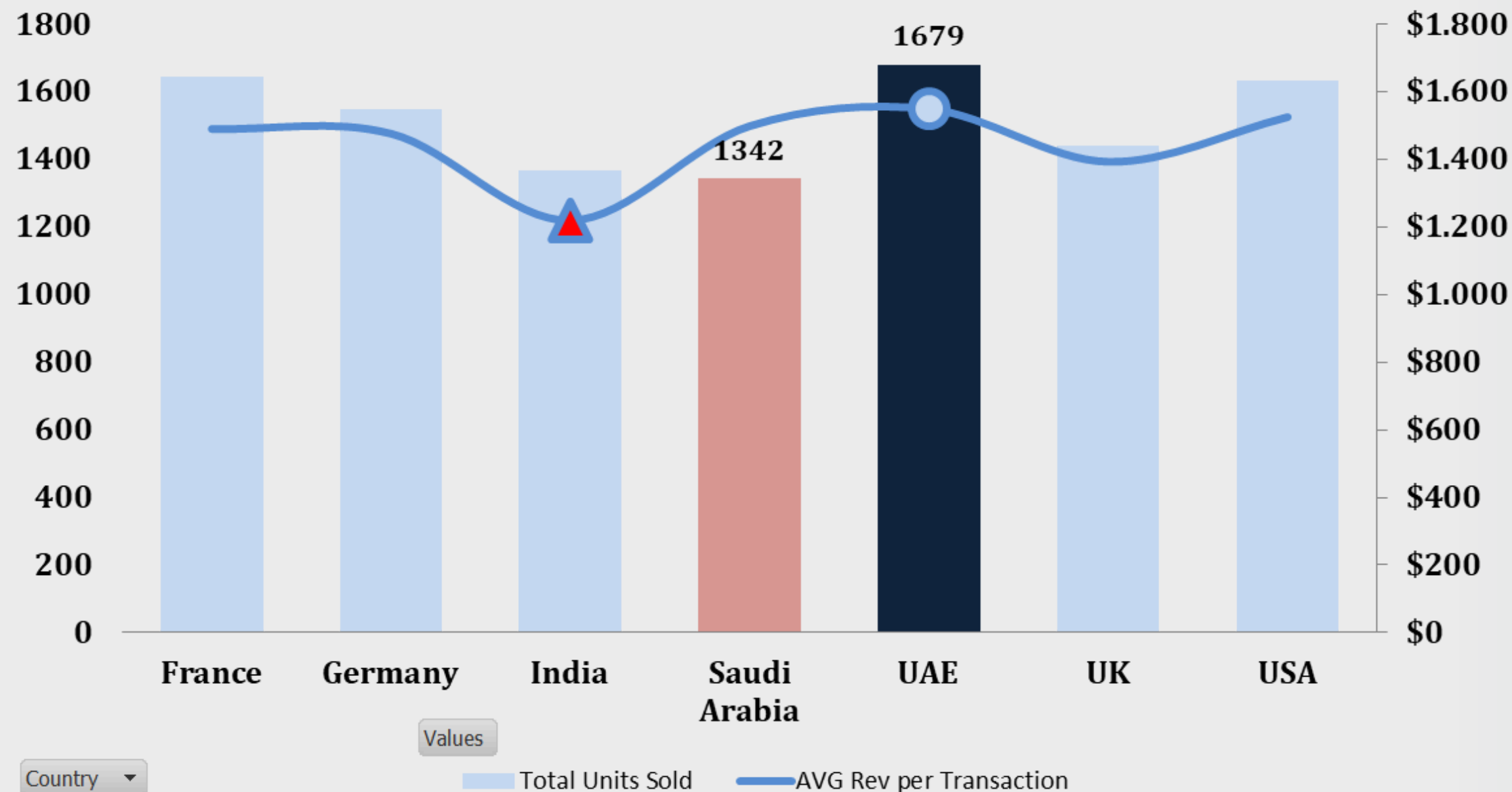
Which country generated the highest total revenue?



“The UAE generated the highest revenue among all countries”

The revenue contribution was only slightly higher than that of the other countries, ranging from 12.19% to 16.44%. This indicates a well-balanced market portfolio, where overall **business performance is supported by multiple markets rather than relying heavily on a single country**

How does sales performance trend across each country?



- **UAE recorded the highest values across key sales metrics**, including revenue, transaction volume, units sold, and revenue efficiency metrics. However, the differences compared with the other markets were relatively small, indicating a broadly similar level of performance across countries
- **Saudi Arabia** achieved strong revenue efficiency despite lower sales volume, **suggesting higher customer spending per transaction**
- In contrast, **India** recorded the lowest average revenue per transaction and the lowest total units sold. **indicating weaker customer demand and lower purchasing value compared to other markets**

Despite the similar overall performance, the underlying revenue drivers varied across markets. Different combinations of transaction volume, units sold, and average transaction value contributed to revenue generation, suggesting distinct customer purchasing patterns in each market

How does sales performance trend across each channel?

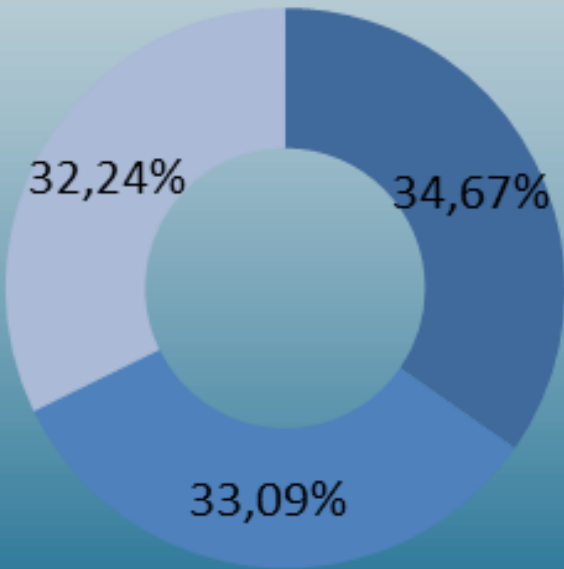
Channel	Total Revenue	Total Units Sold	Total Transaction
Retail Store	\$506,1K	3597	339
Mall	\$483,1K	3533	333
Online	\$470,7K	3521	334

“Retail store generated the highest revenue among all sales channels”

- One-Way ANOVA was conducted to test whether revenue differences among channels are statistically significant.

p-value = 0.746 (>0.05)

Percent Revenue Contribution

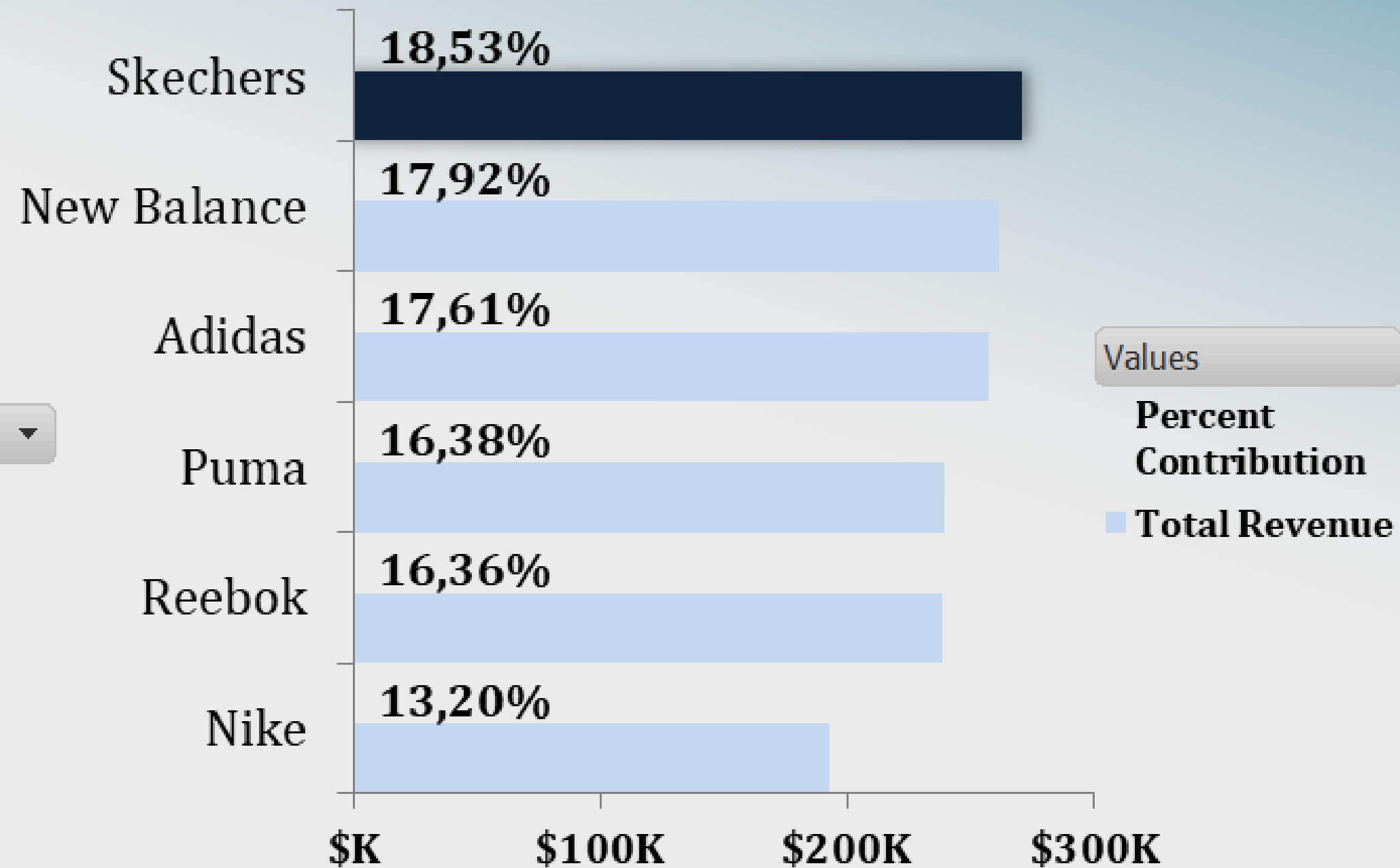


Sales_Channel ▾

■ Retail Store ■ Mall ■ Online

Retail stores generated the highest revenue among all sales channels, although the performance gap across channels was relatively small. indicating that **all channels contributed similarly to overall revenue performance.**

Which Brand Generated the Highest Revenue?



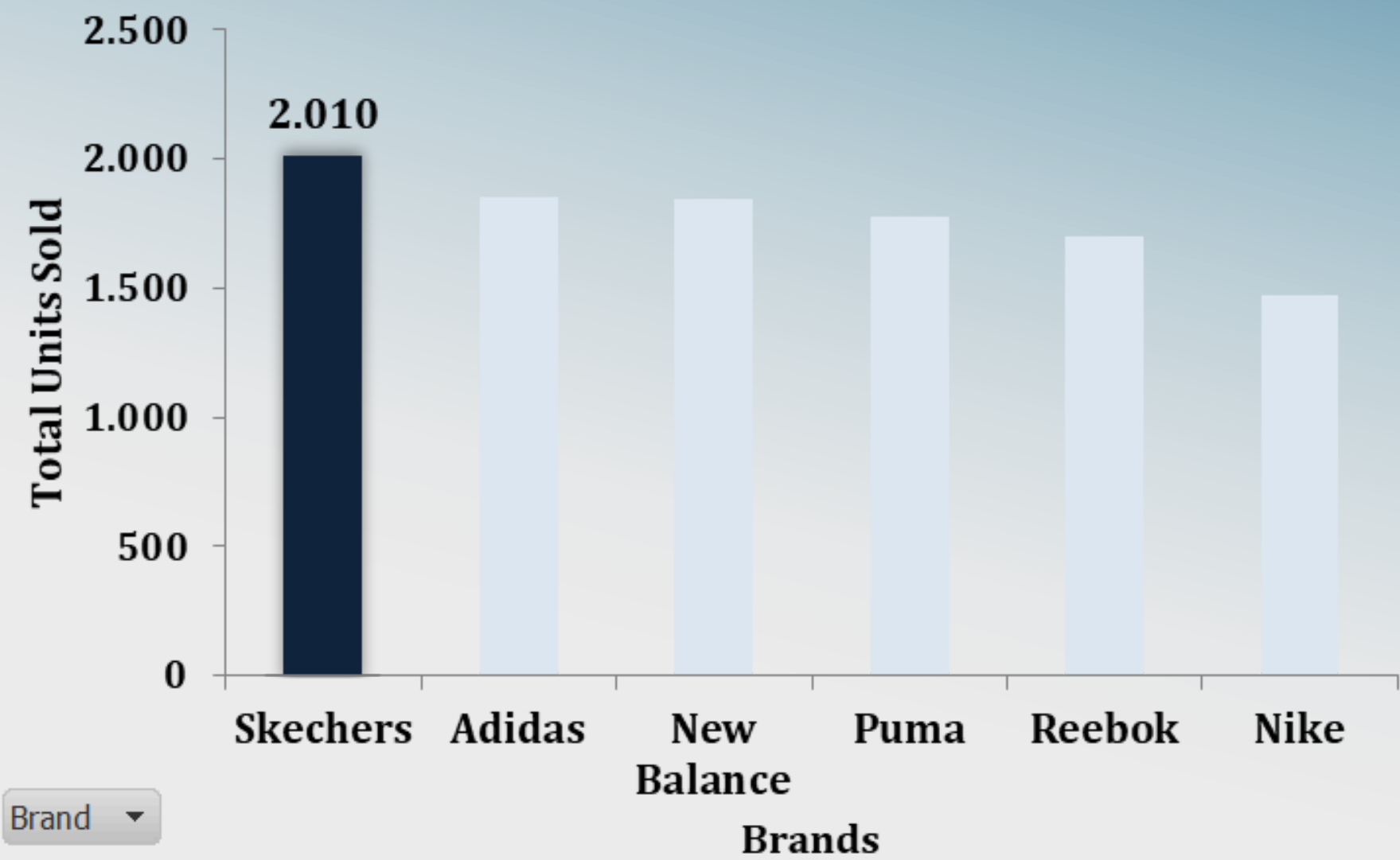
“Skechers generated the highest revenue among all brands”

- One-Way ANOVA was conducted to test whether revenue differences among channels are statistically significant.

p-value = 0,2294 (>0.05)

The revenue contribution across brands is relatively balanced, with the top-performing brand contributing 18% of total revenue. However, the small gap between the highest and lowest contributors suggests that revenue is distributed relatively evenly across brands. **This indicates a well-balanced brand portfolio, where overall business performance is supported by multiple brands rather than relying heavily on a single brand.**

How does sales performance trend across each brand?



Brands	AVG per Product
New Balance	\$142
Reebok	\$140
Adidas	\$139
Puma	\$135
Skechers	\$135
Nike	\$131

“Most brands generated revenue primarily through sales volume, with higher units sold becoming the main driver of revenue performance.”

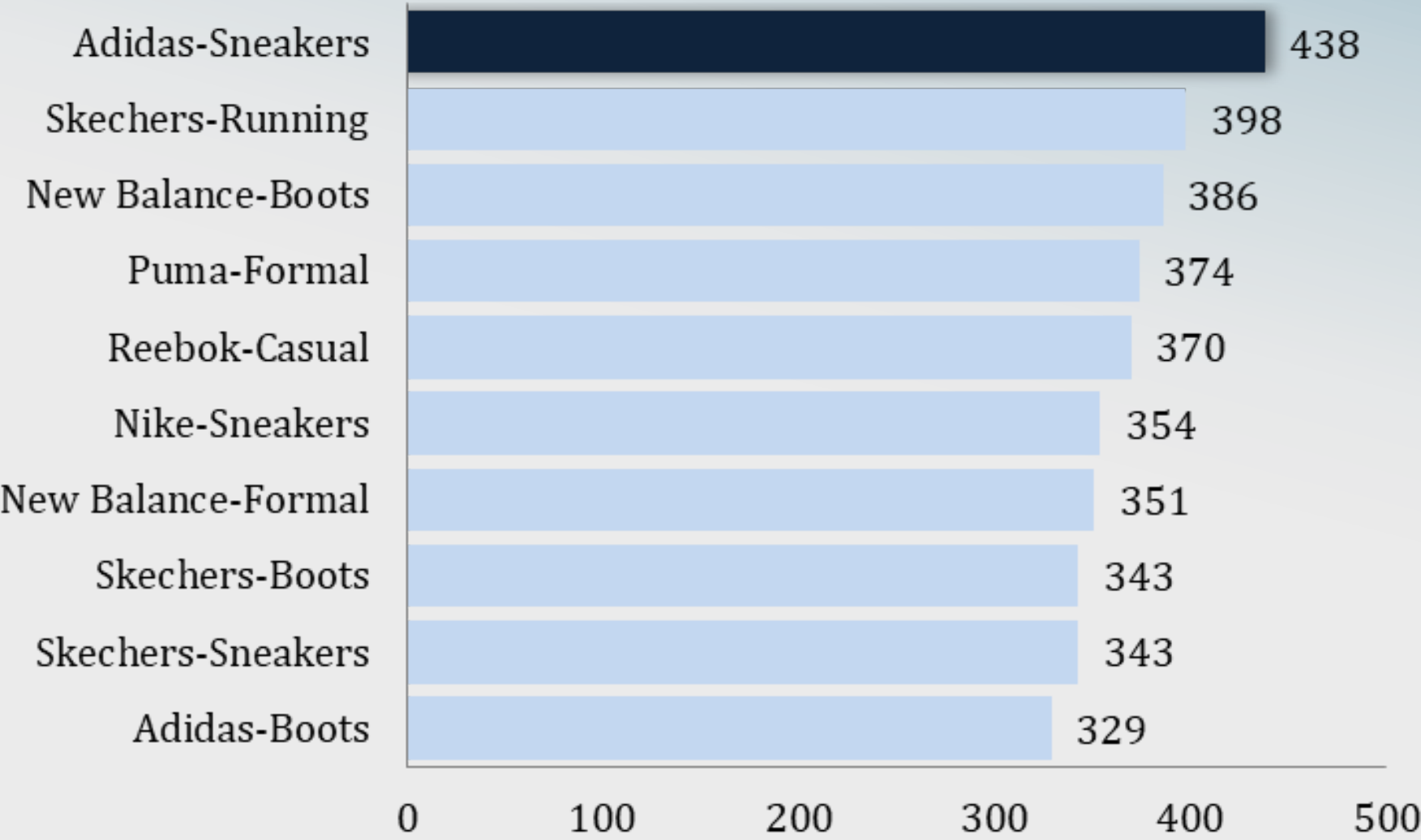
Skechers recorded **the highest revenue** among all brands, supported by its strong sales volume despite having a relatively lower average product value

Nike suggesting weaker overall sales performance compared to other brands.

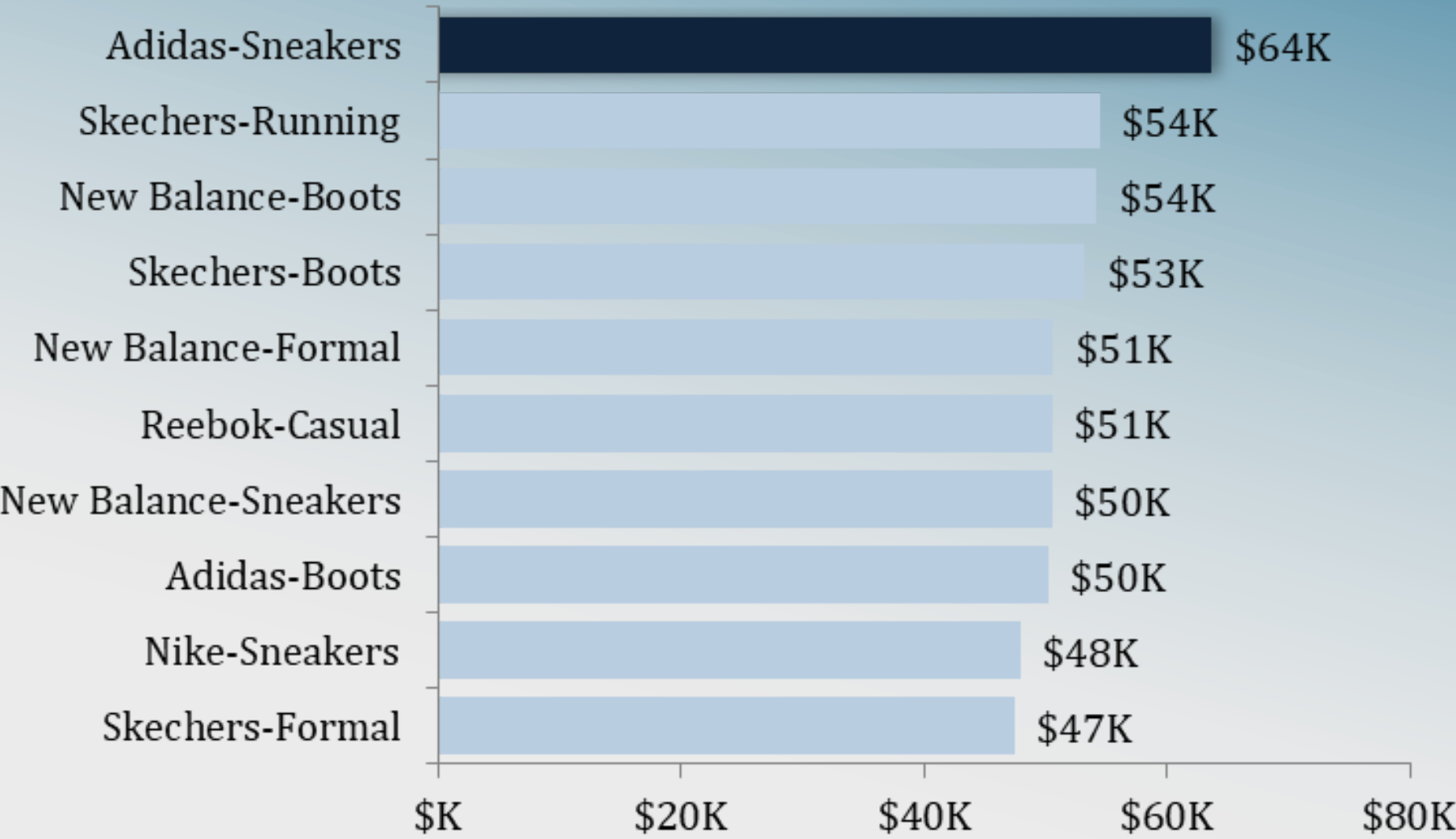
New Balance showed a different pattern, achieving strong revenue contribution through a combination of high unit sales and higher average product value

Which Products Were Purchased the Most and Generated the Highest Revenue?

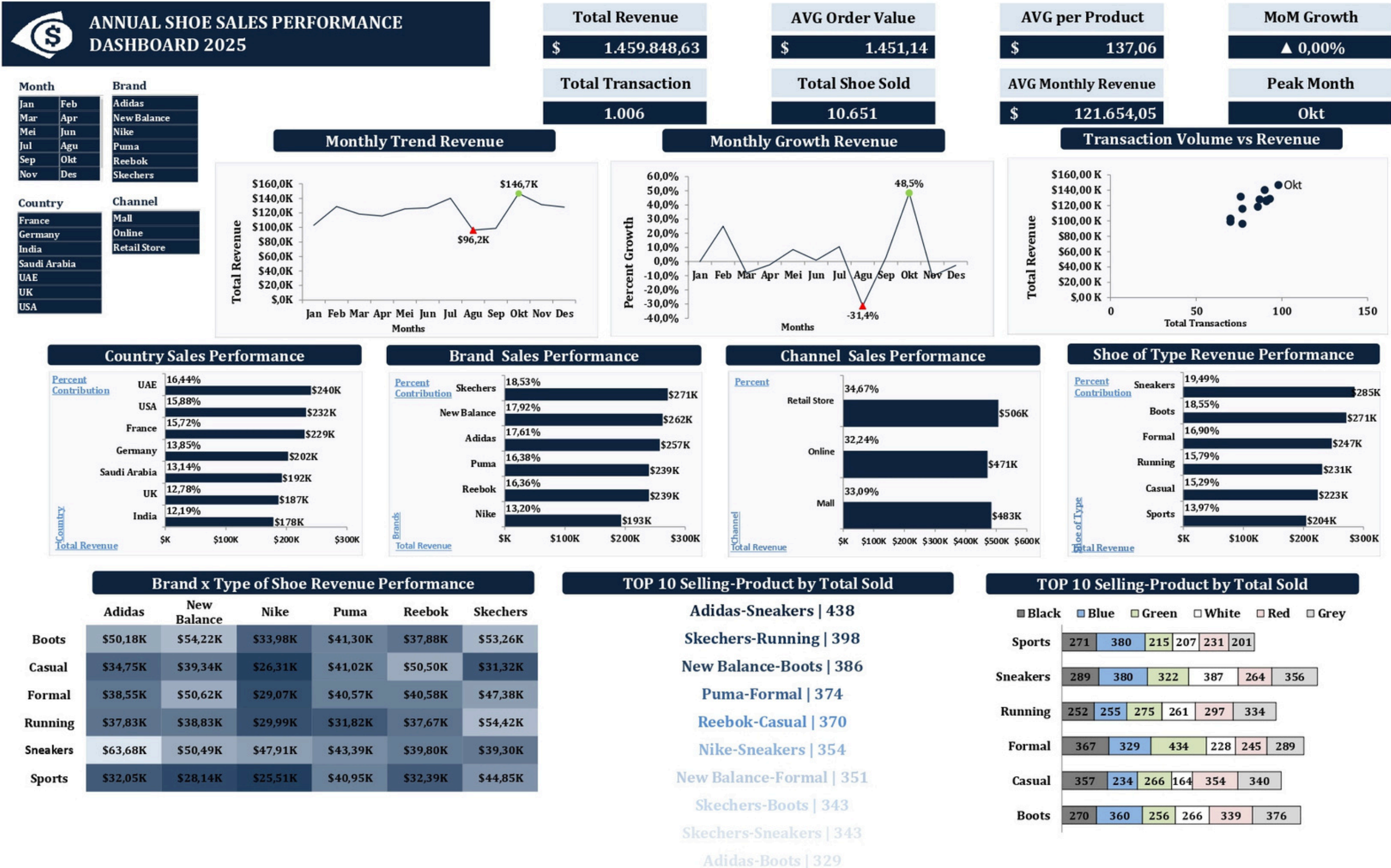
Top 10 Best-Selling Products by Units Sold



TOP 10 Poducts by Total Revenue



DASHBOARD REPORT



Key Insights Driving Revenue Performance and Growth Opportunities

Revenue Performance Shows Monthly Fluctuation

Revenue performance fluctuated throughout the year, with October achieving the highest revenue while August recorded the lowest performance

Channel and Brand Contributions are Relatively Balanced

Revenue contribution was distributed relatively evenly across sales channels and brands, with no single category dominating performance

Regional Performance Differs Mainly Due to Sales Volume and Transaction Value

Market performance varies across countries, with UAE generating the highest revenue contribution while India recorded the lowest performance

Sales Volume is the Main Revenue Driver

Revenue is strongly associated with sales volume, where higher units sold generally resulted in higher revenue performance. Revenue strongly correlates with sales volume ($R^2 = 83\%$)

Turning Insights into Actionable Business Strategies

Focus Area : Q3 Low-Season Recovery (August Revenue Decline)

Strategic Objective : Recover sales volume decline during low-performing periods by stimulating customer demand.

Strategic Action :

Launch a targeted Back-to-School / Late Summer campaign before the historical August revenue decline to stimulate customer demand.

Product Focus :

Prioritize promotional efforts on historically high-performing SKUs, particularly Adidas-Sneakers, while expanding complementary products through bundle promotions.

Offer Mechanism :

Test a bundle promotion (ex : Buy 1, Get 20% Off Second Pair) to encourage multi-item purchases and increase units per transaction

Turning Insights into Actionable Business Strategies

Focus Area : Regional Market Optimization

Strategic Objective : Optimize regional market performance through market specific growth strategies

Strategic Action :

India :

- Launch localized marketing campaigns featuring historically high-performing products to increase customer demand and purchase frequency.
- Test cross-selling bundle offers (ex : Footwear + Accessories or Shoe Care Kits) to encourage higher transaction values.
- Review product assortment and pricing strategy to better align with local customer purchasing behavior

UAE & Saudi Arabia :

- Prioritize premium and higher-priced product offerings in UAE and Saudi Arabia to align with stronger customer purchasing power.
- Ensure adequate availability of top-performing and premium SKUs during high-demand periods to prevent missed sales opportunities.
- Introduce exclusive product collections or limited-time launches to encourage repeat purchases from high-value customers.

Turning Insights into Actionable Business Strategies

Focus Area : Omnichannel Efficiency Strategy

Strategic Objective : Capitalize on balanced channel distribution while maximizing profitability

Strategic Action :

Maintain Balanced Channel Stocking :

Maintain a balanced product assortment across Online, Mall, and Retail channels to support consistent revenue performance

Monthly Multi Metric Performance Tracking :

Monitor channel performance regularly using Revenue, Units Sold, and Transactions to identify changes in customer purchasing patterns

Channel Tailored Promotional Testing:

Optimize channel specific promotional activities based on historical sales performance.

Turning Insights into Actionable Business Strategies

Focus Area : Q4 Peak Season Capitalization

Strategic Objective : Maximize high spending consumer behavior in October–November.

Strategic Action :

Prepare top-performing product availability before peak periods by monitoring historically high-performing product combinations and ensuring sufficient product availability to capture customer demand.

Implement cross-selling and upselling strategies during Q4 (ex : footwear + accessories bundles) to increase units per transaction and Average Revenue per Transaction.